XI The Method of Sampling

一个关于Sampling的基本问题：求$\mathbb{E}[f(z)] = \int f(z)p(z)dz$ （e.g. 求不均匀物体的质量）

1. Rejection sampling

问题：如何从$p(z)$中采样

$p(z)$: distribution hard to sample but easy to evaluate, e.g. Gamma distribution（易于求值，难于采样）

$$p(z) = \frac{1}{Z_p}\tilde{p}(z) = \frac{1}{\int \tilde{p}(z)dz}\tilde{p}(z)$$

$q(z)$: distribution easier to sample

① Generate a number z_0 from the distribution $q(z)$

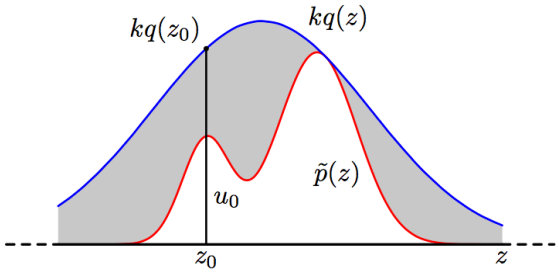
② Generate a number u_0 from the uniform distribution over $[0, kq(z_0)]$

③ if $u_0 > \tilde{p}(z_0)$:

z_0 is rejected

else:

z_0 is retained



（在z轴上投点，符合标准的点的分布即为目标分布；类似于蒙特卡洛投点求面积的过程）

（在$\tilde{p}(z)$越大的地方，点越容易符合标准，密度越大）

Notes: $P(\text{accept} \mid z = z_0) = \frac{\tilde{p}(z_0)}{kq(z_0)}$

$$\Rightarrow P(\text{draw available } z = z_0) = P(\text{draw } z = z_0 \text{ from } q(z)) \cdot P(\text{accept} \mid z = z_0) = \frac{\tilde{p}(z_0)}{k}$$

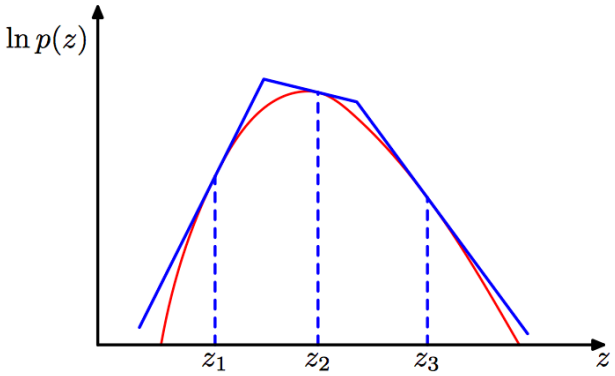
Notes: $P(\text{accept}) = \int \frac{\tilde{p}(z) - 0}{kq(z) - 0} \cdot q(z)dz = \frac{1}{k}\int \tilde{p}(z)dz = \frac{Z}{k}$

显然，k越小，accept概率越大（在高维问题中，accept概率会急速下降）

Notes: $p_s(z) = \frac{p_s(z, \text{accept})}{p_s(\text{accept})} = \frac{\tilde{p}(z)/k}{Z/k} = \frac{\tilde{p}(z)}{Z} = p(z)$, s : sample

Rejection sampling需要一个合适的proposal distribution $q(z)$，能较好地包围$\tilde{p}(z)$以获得较高的accept rate。同样地，这在multimodal和sharply peak经常出现的高维问题中，也是比较难以寻找的。

e.g. Adaptive rejection sampling



2. Importance sampling

与Rejection sampling不同，Importance sampling没有从$p(z)$中取样；

对于$\mathbb{E}[f(z)] = \int f(z)p(z)dz$的求解

为了提高计算效率，我们希望尽可能包括$f(z)p(z)$较大的点

利用proposal distribution $q(z)$,

$$\mathbb{E}[f] = \int f(z)p(z)dz = \int f(z)\frac{p(z)}{q(z)}q(z)dz \approx \frac{1}{L}\sum_{l=1}^{L}\frac{p(z^{(l)})}{q(z^{(l)})}f(z^{(l)})$$

记$r_l = \frac{p(z^{(l)})}{q(z^{(l)})}$为importance weights （用于修正从错误分布$q(z)$取样的误差）

3. Markov Chain Monte Carlo

和Rejection sampling类似，MCMC需要有一个proposal distribution $q(z)$

MCMC的关键在于通过马尔可夫链获得稳态分布$p(z)$

① Generate a number $z^{(\tau)}$ from the distribution $q(z)$

② Generate a candidate z^* from the distribution $q(z)$

③ Accept z^* under some criterion $\Rightarrow z^{(\tau+1)} = \cdots$

e.g. basic Metropolis algorithm

$$A(z^*, z^{(\tau)}) = \min\left(1, \frac{\tilde{p}(z^*)}{\tilde{p}(z^{(\tau)})}\right)$$

Generate a number u from the uniform distribution over (0,1)

if $u < A(z^*, z^{(\tau)})$:

accept z^*, $z^{(\tau+1)} = z^*$

else:

discard z^*, $z^{(\tau+1)} = z^{(\tau)}$

Notes: 要求$q(z)$满足$q(A \mid B) = q(B \mid A)$

Some basic knowledge of Markov Chain:

A first-order Markov chain is defined to be **a series of random variables** $z^{(1)}, z^{(2)}, \ldots, z^{(m)}$

(stochastic process)

s.t. $p(z^{(m+1)} \mid z^{(1)}, \ldots z^{(m)}) = p(z^{(m+1)} \mid z^{(m)})$

The transition probabilities is defined as $T_m(z^{(m)}, z^{(m+1)}) := p(z^{(m+1)} \mid z^{(m)})$

A distribution is said to be invariant w.r.t. a Markov Chain if each step in the chain leaves that **distribution** invariant. $\Rightarrow p^*(z) = \sum_{z'} T(z', z)p^*(z')$

$p^*(z)T(z, z') = p^*(z')T(z', z) \Rightarrow$ the distribution is invariant

e.g. The Metropolis-Hastings algorithm

$$A(z^*, z^{(\tau)}) = \min\left(1, \frac{\tilde{p}(z^*)q_k(z^{(\tau)} \mid z^*)}{\tilde{p}(z^{(\tau)})q_k(z^* \mid z^{(\tau)})}\right)$$

Generate a number u from the uniform distribution over (0,1)

if $u < A(z^*, z^{(\tau)})$:

accept z^*, $z^{(\tau+1)} = z^*$

else:

discard z^*, $z^{(\tau+1)} = z^{(\tau)}$

可以验证：

$p(z)q_k(z \mid z')A_k(z, z') = p(z')q_k(z' \mid z)A_k(z', z) \Rightarrow p(z)$ is invariant

Notes: 推导该算法中$A(z^*, z^{(\tau)})$

由$p^*(z)T(z, z') = p^*(z')T(z', z)$知

$\frac{T(z, z')}{T(z', z)} = \frac{p^*(z')}{p^*(z)}$，而$T(z, z') = Q(z' \mid z)A(z', z), T(z', z) = Q(z \mid z')A(z, z')$

$$\Rightarrow \frac{Q(z' \mid z)A(z', z)}{Q(z \mid z')A(z, z')} = \frac{p^*(z')}{p^*(z)} \Rightarrow A(z', z) = \frac{p^*(z')Q(z \mid z')}{p^*(z)Q(z' \mid z)}A(z', z)$$

令$t(z', z) = \frac{p^*(z')Q(z \mid z')}{p^*(z)Q(z' \mid z)}$，可知$t(z, z') = \frac{1}{t(z', z)}$

而$A(z', z) = t(z', z)A(z, z') \triangleq F(t(z', z))$

由于$A(z, z') = t(z, z')A(z', z) = \frac{1}{t(z', z)}A(z', z) \Rightarrow F(x) = F(1/x)/x$

故取$F(x) = \min\{1, x\}$即可

e.g. Gibbs sampling

Gibbs sampling可以看成是Metropolis-Hastings algorithm的特殊情况

将更新得到$z_k^{(\tau+1)}$后的$\boldsymbol{z}$记为z^*，则有$q_k(z^* \mid z) = p(z_k \mid z_{\text{except } k})$

结果是：总是接受z^*（$\frac{\tilde{p}(z^*)q_k(z^{(\tau)} \mid z^*)}{\tilde{p}(z^{(\tau)})q_k(z^* \mid z^{(\tau)})} = \frac{p(z_k^* \mid z_{\text{except } k}^*)p(z_{\text{except } k}^*)p(z_k^{(\tau)} \mid z_{\text{except } k}^*)}{p(z_k^{(\tau)} \mid z_{\text{except } k}^{(\tau)})p(z_{\text{except } k}^{(\tau)})p(z_k^* \mid z_{\text{except } k}^{(\tau)})} = 1$）

1. Initialize $\{z_i : i = 1, \ldots, M\}$
2. For $\tau = 1, \ldots, T$:
   - Sample $z_1^{(\tau+1)} \sim p(z_1 | z_2^{(\tau)}, z_3^{(\tau)}, \ldots, z_M^{(\tau)})$.
   - Sample $z_2^{(\tau+1)} \sim p(z_2 | z_1^{(\tau+1)}, z_3^{(\tau)}, \ldots, z_M^{(\tau)})$.

     $\vdots$
   - Sample $z_j^{(\tau+1)} \sim p(z_j | z_1^{(\tau+1)}, \ldots, z_{j-1}^{(\tau+1)}, z_{j+1}^{(\tau)}, \ldots, z_M^{(\tau)})$.

     $\vdots$
   - Sample $z_M^{(\tau+1)} \sim p(z_M | z_1^{(\tau+1)}, z_2^{(\tau+1)}, \ldots, z_{M-1}^{(\tau+1)})$.